Dataset link :

https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews

Assignment: Sentiment Analysis on IMDB Movie Reviews

Course: Natural Language Processing / Machine Learning

Objective:
In this assignment, you will build a complete text classification pipeline to classify movie reviews as **Positive** or **Negative** using the IMDB Movie Review Dataset. You will apply the text preprocessing techniques learned in class and compare multiple machine learning algorithms.

Learning Outcomes

After completing this assignment, students will be able to

- Perform text preprocessing
- Convert text into numerical features
- Train multiple machine learning classifiers
- Evaluate classification models
- Interpret performance metrics
- Compare different text representations

Dataset

Use the IMDB Movie Reviews Dataset.

You may use either

- Kaggle IMDB Dataset
- OR TensorFlow IMDB Dataset

Dataset contains

- Review Text
- Sentiment (Positive / Negative)

Approximately **50,000** movie reviews.

Part 1: Import Libraries

Import all necessary libraries.

Example libraries include

- pandas
- numpy
- matplotlib
- nltk
- re
- sklearn

Part 2: Load the Dataset

Load the dataset into a pandas DataFrame.

Tasks

1. Display first five rows.
2. Display dataset shape.
3. Display column names.
4. Check data types.

Part 3: Exploratory Data Analysis

Perform basic EDA.

Tasks

- Number of positive reviews
- Number of negative reviews
- Check missing values
- Check duplicate rows
- Average review length

- Longest review
- Shortest review

Visualization

Create

- Sentiment distribution bar plot
- Histogram of review lengths

Questions

1. Is the dataset balanced?
2. What is the average review length?

Part 4: Text Cleaning

Using techniques learned in class, clean the review text.

Perform the following

✓ Convert to lowercase

✓ Remove HTML tags

✓ Remove URLs

✓ Remove punctuation

✓ Remove numbers

✓ Remove chat abbreviations

✓ Remove stopwords

✓ Remove emojis

✓ Correct spelling (optional)

✓ Lemmatization

Create a new column

`clean_review`

Part 5 : Feature Engineering from Text –

(Create new numerical features from the cleaned review text.)

Task 1: Review Length

Create a feature representing the total number of words in each review.

Task 2: Character Count

Calculate the total number of characters.

example	Character Count
Great movie	11

Task 3: Average Word Length

Average Word Length=Number of Words / Total Characters

Task 4: Sentence Count

Count the number of sentences in each review.

Task 5: Average Sentence Length

Average Sentence Length = Number of Sentences/ Total Words

Task 6: Unique Word Count

`movie movie movie good ;` Unique words = 2

Task 7: Lexical Diversity

Lexical Diversity =Total Words/ Unique Words

This measures vocabulary richness.

Visualization

Plot the distribution of

- Review Length
- Character Count
- Lexical Diversity
- Average Sentence Length

using histograms or boxplots.

Statistical Comparison

Compare these features for positive and negative reviews.

Example

Feature	Positive Mean	Negative Mean
Review Length		
Avg Word Length		
Character Count		
Lexical Diversity		

Analysis Questions

1. Do positive reviews tend to be longer than negative reviews?
2. Which feature shows the largest difference between classes?
3. Which handcrafted feature do you think would be most useful for classification?
4. Which feature appears least useful?

Write a paragraph of your analysis - What you tried to understand from this features